## NO.3 BRAKE SYSTEM WARNING LIGHT REMAINS ILLUMINATED [DYNAMIC STABILITY CONTROL (DSC)]

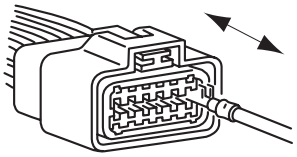
id0403b2897100

| 3 | Brake system warning light remains illuminated |
|---|---|
| POSSIBLE CAUSE | • Brake fluid amount is low.<br>• Parking brake does not release.<br>• Parking brake switch signal error input to instrument cluster.<br>• Brake fluid level sensor signal error input to body control module (BCM).<br>• DSC HU/CM detected malfunction. (Input and output device malfunction)<br>• No connection at DSC HU/CM connector (When DSC HU/CM connector is removed, ABS warning light, brake system warning light, DSC indicator light, and DSC OFF indicator light are illuminated.)<br>• DSC HU/CM detects low voltage in power supply.<br>• DSC HU/CM ground malfunction (When DSC HU/CM ground is not securely connected, ABS warning light, brake system warning light, DSC indicator light, and DSC OFF indicator light are illuminated but DTCs are not display.)<br>• Instrument cluster internal malfunction<br>• DSC HU/CM does not operate. |

### Diagnostic procedure

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **INSPECT BRAKE FLUID AMOUNT AND VERIFY THAT PARKING BRAKE IS RELEASED**<br>• Inspect the brake fluid amount and verify that the parking brake is released.<br>• Is the brake fluid amount normal?<br>• Is the parking brake lever released? | Yes | Go to the next step. |
| | | No | Add the brake fluid or release the parking brake lever.<br>If the brake fluid amount is less than specification:<br>• Inspect and repair the brake line for leakage. |
| 2 | **CONFIRM DSC HU/CM DTC**<br>• Retrieve the DSC HU/CM DTC using the M-MDS.<br>(See DTC INSPECTION [DSC HU/CM].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [DSC HU/CM].) |
| | | No | Go to the next step. |
| 3 | **CONFIRM INSTRUMENT CLUSTER OPERATION RECORD**<br>• Retrieve the warning system operation history using the M-MDS.<br>(See INSTRUMENT CLUSTER OPERATION RECORD.)<br>• Is the brake system warning light illumination history recorded? | Yes | If the Brake Warning Lamp is recorded:<br>• Go to the next step.<br>If the Brake Warning Lamp (Brake fluid low) is recorded:<br>• Go to Step 5.<br>If the Brake Warning Lamp (Parking Brake ON over 20 km/h or 12.4 mile/h) is recorded:<br>• Go to Step 7. |
| | | No | Go to Step 9. |
| 4 | **VERIFY IF MALFUNCTION IS IN INSTRUMENT CLUSTER OR ELSEWHERE**<br>• Perform Warning/Indicator Light Illumination Inspection.<br>(See INSTRUMENT CLUSTER INSPECTION.)<br>• Does the brake system warning light illumination?. | Yes | Go to Step 9. |
| | | No | Replace the instrument cluster.<br>(See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.) |
| 5 | **VERIFY IF MALFUNCTION IS IN BRAKE FLUID LEVEL SENSOR OR ELSEWHERE**<br>• Inspect the brake fluid level sensor.<br>(See BRAKE FLUID LEVEL SENSOR INSPECTION.)<br>• Is the brake fluid level sensor normal? | Yes | Go to the next step. |
| | | No | Replace the master cylinder assembly.<br>(See MASTER CYLINDER REMOVAL/INSTALLATION [L.H.D.].)<br>(See MASTER CYLINDER REMOVAL/INSTALLATION [R.H.D.].) |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| *6 | **INSPECT FOR SHORT TO GROUND IN WIRING HARNESS BETWEEN BODY CONTROL MODULE (BCM) AND BRAKE FLUID LEVEL SENSOR**<br>• Inspect for short to ground in the following circuit:<br>— Between brake fluid level sensor and body control module (BCM) terminal 3R<br>• Is a short to ground detected? | Yes | Repair or replace the wiring harness for a possible short to ground. |
| | | No | Retrieve the body control module (BCM) DTC using the M-MDS.<br>(See DTC INSPECTION [BLIND SPOT MONITORING (BSM) SYSTEM].)<br>If the DTC remains:<br>• Go to the applicable DTC inspection.<br>(See DTC TABLE [BLIND SPOT MONITORING (BSM) SYSTEM].)<br>If the DTC does not remain:<br>• Replace the body control module (BCM).<br>(See BODY CONTROL MODULE (BCM) REMOVAL/ INSTALLATION.) |
| 7 | **INSPECT WHETHER MALFUNCTION IS IN PARKING BRAKE SWITCH OR ELSEWHERE**<br>• Inspect the following parts for continuity:<br>— Parking brake switch<br>(See PARKING BRAKE SWITCH INSPECTION.)<br>• Is the continuity condition normal? | Yes | Go to the next step. |
| | | No | Replace the parking brake.<br>(See PARKING BRAKE LEVER REMOVAL/ INSTALLATION.) |
| *8 | **INSPECT FOR SHORT TO GROUND IN WIRING HARNESS BETWEEN INSTRUMENT CLUSTER AND PARKING BRAKE SWITCH**<br>• Inspect for short to ground in the following circuit:<br>— Between parking brake switch and instrument cluster terminal 1X<br>• Is the short to ground detected? | Yes | Repair or replace the wiring harness for a possible short to ground. |
| | | No | Retrieve the instrument cluster DTC using the M-MDS.<br>(See DTC INSPECTION [INSTRUMENT CLUSTER].)<br>If the DTC remains:<br>• Go to the applicable DTC inspection.<br>(See DTC TABLE [INSTRUMENT CLUSTER].)<br>If the DTC does not remain:<br>• Replace the instrument cluster.<br>(See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.) |
| 9 | **VERIFY IF MALFUNCTION IS IN DSC HU/CM CONNECTOR FOR POOR CONNECTION OR ELSEWHERE**<br>• Switch the ignition ON.<br>• Does the ABS warning light, brake warning light, TCS/DSC indicator light, and DSC OFF indicator light turn off after 2 to 4 s? | Yes | Inspect the DSC HU/CM connector or terminal. (poor connection intermittently)<br>• If there is a malfunction, repair or replace any malfunctioning parts according to the inspection result. |
| | | No | Go to the next step. |
| 10 | **INSPECT BATTERY VOLTAGE**<br>• Switch the ignition off.<br>• Inspect the battery voltage.<br>(See BATTERY INSPECTION.)<br>• Is the battery voltage normal? | Yes | Go to the next step. |
| | | No | Recharge the battery and inspect the charging system.<br>(See BATTERY INSPECTION.)<br>(See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)<br>(See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)<br>(See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)<br>• If there is a malfunction, repair or replace any malfunctioning parts according to the inspection result. |
| 11 | **VERIFY IF MALFUNCTION IS IN CHARGING SYSTEM OR ELSEWHERE**<br>• Measure the battery voltage with electrical loads (such as A/C, headlight) applied while idling.<br>• Is the battery voltage normal? | Yes | Go to the next step. |
| | | No | Inspect the charging system (drive belt, generator).<br>(See DRIVE BELT INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)<br>(See DRIVE BELT INSPECTION [SKYACTIV-D 1.5].)<br>(See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)<br>(See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)<br>• If there is a malfunction, repair or replace any malfunctioning parts according to the inspection result. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 12 | **INSPECT CONNECTION OF DSC HU/CM CONNECTOR**<br>• Inspect connection of the DSC HU/CM connector.<br>• Is the DSC HU/CM connector connected securely? | Yes | Go to the next step. |
| | | No | Connect the DSC HU/CM connector securely, then go to the next step. |
| 13 | **INSPECT CONNECTION OF CONNECTOR TERMINAL**<br>• Is there a poor connection in the DSC HU/CM, instrument cluster, or related-connector terminal?<br>  <br>INSERT MALE TERMINAL INTO MATCHING SHAPE OF FEMALE TERMINAL AND CHECK FOR EXCESSIVE PLAY | Yes | Repair the poor connection location of the terminal. |
| | | No | Go to the next step. |
| 14 | **VERIFY IF MALFUNCTION IS IN WIRING HARNESS (BETWEEN DSC HU/CM POWER SUPPLY AND DSC HU/CM FOR CONTINUITY) OR ELSEWHERE**<br>• Switch the ignition ON.<br>• Measure the voltage at the DSC HU/CM terminal Q (wiring harness-side).<br>• Is the voltage approx. 12 V? | Yes | Go to the next step. |
| | | No | Refer to the wiring diagram and verify whether or not there is a common connector between DSC HU/CM terminal Q (wiring harness-side) and ignition switch.<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has an open circuit. |
| 15 | **VERIFY IF MALFUNCTION IS IN WIRING HARNESS (BETWEEN DSC HU/CM AND GROUND FOR CONTINUITY) OR ELSEWHERE**<br>• Switch the ignition off.<br>• Disconnect the DSC HU/CM connector.<br>• Inspect for continuity between DSC HU/CM terminal AL (wiring harness-side) and body ground.<br>• Is there continuity? | Yes | Replace the DSC HU/CM. (open circuit in the DSC HU/CM)<br>(See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].)<br>(See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].) |
| | | No | Repair or replace the wiring harness for a possible open circuit and poor contact at the ground point. |
| 16 | **CONFIRM INSTRUMENT CLUSTER DTC**<br>• Retrieve the instrument cluster DTC using the M-MDS.<br>(See DTC INSPECTION [INSTRUMENT CLUSTER].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [INSTRUMENT CLUSTER].) |
| | | No | Replace the instrument cluster.<br>(See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.) |

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while doing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, verify that connectors, terminals and wiring harnesses are connected correctly and undamaged.